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BRAC UNIVERSITY

CSE360: Computer Interfacing

Lab Project Report

Title: *Smart Bike Security and Tamper Detection System*

by

Group_No: 07

Section: 08

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Abstract

The project presents a Smart Security System for motorcycles, with the aim to provide robust, low powered security solutions in order to address limitations of traditional mechanical locks and basic electronic alarm systems. For the methodology, we incorporate Arduino uno as the central controller with ADXL345 accelerometer for motion detection, reed switch for forced entry, DS3231 RTC for time stamping, GSM module for live notifications, and GPS module for location services. The system manages two-stage alarms based on vibration and sirens, auto-arm functionality, lock out feature for thwarting brute force attacks, and low power consumption modes like ultra-deep sleep. The notable results from this project includes the effective deployment of the low power consumption mode, proper implementation of interrupt based motion detector, tamper logging through EEPROM and sending SMS messages when required. The broader sustainability considerations focus on low energy consumption.

Keywords

Motorcycle security, Low power consumption design, Embedded systems, Arduino UNO.

1. Introduction

Motorcycle theft remains one of the persistent problems globally. Conventional methods of the use of handle lock, disc lock and basic loud alarms are prone to defeat or are ignored due to high rates of false alarm due to the lack of the intelligence to differentiate between genuine theft and accidental disturbance of a motorcycle, e.g due to wind or other factors. In addition, they do not have any mechanism of sending a warning or tracking of stolen motorbikes.

The objective of the project is to design a smart security system for motorcycles in order to overcome the limitations of traditional systems. The system realises these goals by the utilization of a combination of accelerometer sensor and magnetic field detector (reed switch) to intelligently detect genuine theft, a 2-stage alarm mechanism of silent alarm, followed by loud siren and SMS alerts with GPS tracking capability. The system operates independently with an isolated power source so that even if the motorcycle power source remains off, the system operation works. The battery life is preserved by maintaining ultra low power consumption.

Computer interfacing is highly integrated with this project through the use of various analog and digital sensors like I2C, GPIO, actuators like PWM for buzzer, communication modules like UART for GSM, GPS along with central microcontroller Arduino. Our project relevance can be achieved as the design and programming of communication protocols, interrupt handling, power management logic are core computer interfacing concepts.

2. Related Work/ Inspiration

Several existing systems motivated this project. One study used an Arduino with RFID, GSM and GPS in order to send an SMS with location upon unauthorized access and cut off the engine via a relay, ensuring vehicle security(Purwanto, 2019). Another study approached authentication security with the use of RFID with a fingerprint sensor for biometric verification and used a simple vibration sensor to trigger an alarm(Hidayanti, 2020). Furthermore, another study utilizes a single shock sensor paired with GPS/GSM modules, though it relies on the motorcycle's main battery(Artano, 2020). These projects prove that integration of GSM, GPS with a microcontroller on a cost-effective platform is feasible and effective for theft alerts and engine cut-off. They inspired our work by demonstrating the core building blocks, namely arduino, GPS, GSM and relay.

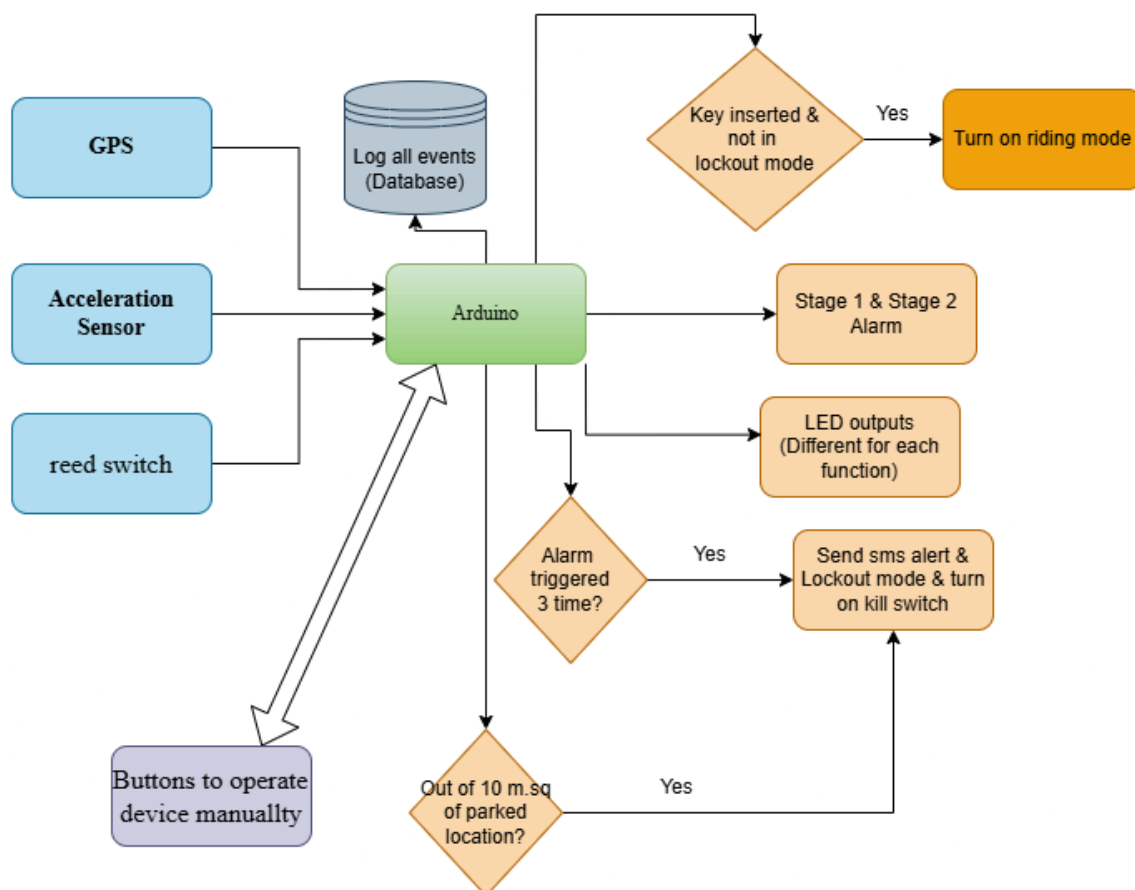
Though our work took inspiration for remote notification and tracking, it stands out by solving some unaddressed issues. Firstly, the project distinguishes itself through addressing the issue of false alarm due to accidental bumps. This is achieved with the use of a sensor fusion using both ADXL345 accelerometer and reed switch to replace simple vibration sensors so accidental bumps can be differentiated from sustained tampering. Secondly, the introduction of a dual-stage alarm which causes soft vibrations for motion less than 3s and triggers a loud siren for continuous motion. A lockout mode is also implemented after three triggers. These features are unique to our project as they are absent in the existing works. Thirdly, an ultra low-power deep sleep mode is implemented using an isolated battery such that the system remains active even if the motorcycle's main battery is disconnected. Lastly, a non-volatile tamper log using EEPROM and RTC timestamps is a unique feature that helps to provide forensic evidence for event logging. Thus, our project fills the gap of false alarm reduction, power autonomy,

anti-brute force logic and event logging and offers a more robust, sustainable and user-friendly solution for motorcycle security.

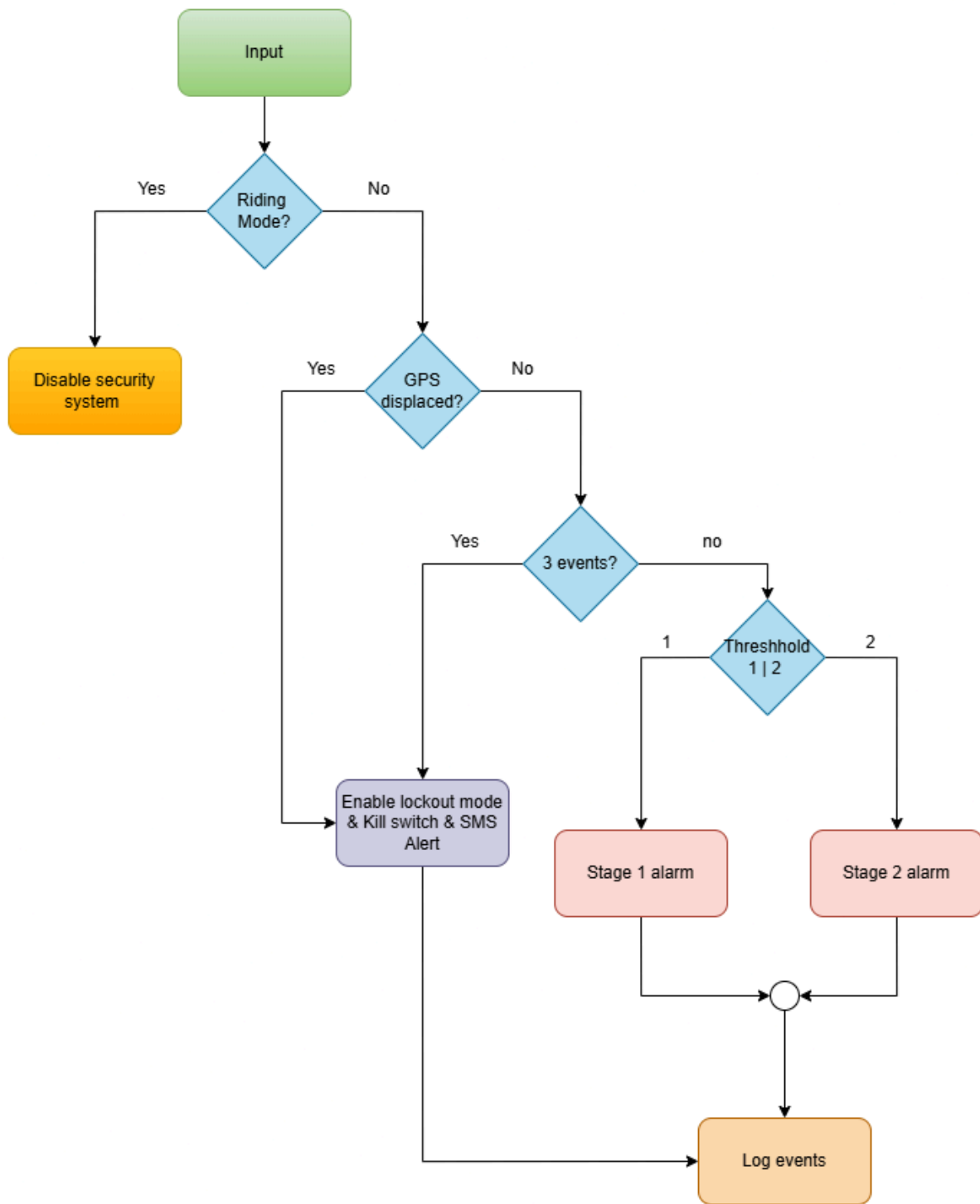
3. Technical Approach

System Architecture:

The system is divided into three core parts. **“Brain”** (Arduino), where all the computation and decision making is done. **“Input”** (gps, acceleration, reed switch), from where the necessary information is coming from for any decision making. And finally **“Feedback”** (buzzer, haptic, LED, sms), which is triggered by the brain when the input values meet a certain threshold.



Block Diagram of the System



WorkFlow Diagram

Components Used:

Hardware List:

1. Arduino Uno: central microcontroller
2. ADXL345 accelerometer: digital accelerometer with programmable interrupt output which is used to detect tampering.
3. DS3231 RTC Module: a real-time clock with I2C interface and battery backup to record tamper log events in EEPROM
4. NEO-6M GPS Module: satellite positioning module used via UART which gives location coordinates for SMS alerts during alarm
5. SIM800L GSM/GPRS Module: GSM/GPRS module for sending SMS text alerts to the owner's phone via UART
6. Reed Switch: Magnetic contact sensor which can indicate forced entry
7. Vibration Motor: a silent alert is established using this small DC motor with an offset weight
8. Active Speaker Buzzer: a 3-pin buzzer to alert as per stages
9. TP4056 Li-ion Charging Module
10. XL6009 DC-DC Boost Module: adjustable DC-DC step-up converter used to boost voltage for stable power.
11. 18650 Li-ion Battery: battery to power the system
12. Battery Holder
13. 5V Relay Module: electromechanical relay driven by a transistor used to implement kill-switch
14. Tactile Push Buttons: switches to provide input for arming, disarming and other codes
15. LEDs: Output light used to detect current mode
16. Resistors
17. Full Size Breadboard
18. Jumper Wires – Male to Male and Male to Female

Software Tools:

Arduino IDE used to implement the system. Different libraries EEPROM, RTCLib.h,

Adafruit_ADXL345_U.h, AT command set etc are used from this library to implement the features.

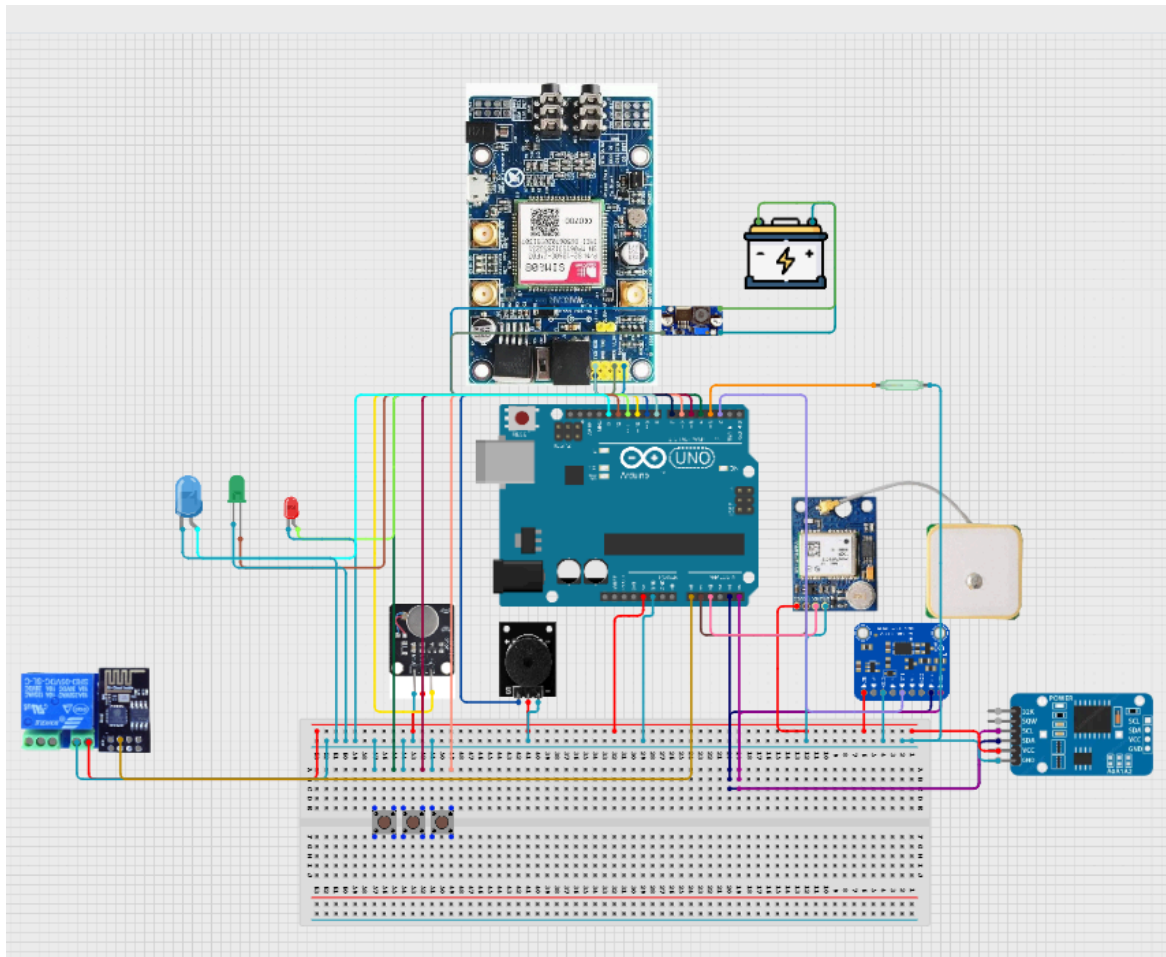
Cost breakdown:

Component	Quantity	Unit price(bdt)	Total (bdt)
Arduino Uno	1	510	510
ADXL345 accelerometer	1	250	250
DS3231 RTC Module	1	244	244
NEO-6M GPS Module	1	990	990
SIM800L GSM/GPRS Module	1	450	450
Reed Switch	1	180	180
Vibration Motor	1	220	220
Active Speaker Buzzer	1	59	59
TP4056 Li-ion Charging Module	1	80	80
XL6009 DC-DC Boost Module	1	166	166
18650 Li-ion Battery	2	300	600
Battery Holder	1	100	100
5V Relay Module	1	50	50
Tactile Push Buttons	3	5	15
LEDs	3	2	6

Resistors	10	1	10
Full Size Breadboard	1	159	159
Jumper Wires – Male to Male	1 set	160	160
Jumper Wires – Male to Female	1 set	150	150
Miscellaneous (soldering wire)			60
Total			4,459

Thus, the project is a highly affordable, feature-rich smart security system.

Circuit/Schematic:



Communication Protocol

We use three main communication methods to link the components. ADXL345 and DS3231 RTC have the same I2C protocol, with pins A4 and A5 connecting them to exchange data and time stamps. The NEO-6M GPS and SIM800L GSM modules are based on UART serial communication, with the SoftwareSerial library generally needed to handle multiple streams of data. Basic parts such as the Reed Switch and Vibration Motor work through the Digital I/O where high or low voltage is used to sense and activate. The GSM module needs its own power source because of the high current during transmission which can cause instability. These protocols combine to enable the Arduino to manage motion sensing, location tracking, and remote alerts easily.

Implementation

Input Layer

- **GPS Module:** It provides real time coordinates information to the Arduino to monitor the bike's position. If the bike is moved out of the set geofence through GSM module sms will be sent to the bike owner.
- **Acceleration Sensor:** It monitors the movement of the bike and sends data to the microcontroller for motion detection.
- **Reed Switch:** It checks if the bike is moved or trying to steal while it is parked.
- **Manual Buttons:** Three buttons are used to allow the user to manually interact with the system. For example, entering a secret sequence to disarm it.

Decision Layer

- **Riding Mode Check:** The arduino checks if the bike key is inserted, in our case we are using Pin D3 as bike key. If the key is grounded that means the key is inserted. If the key is inserted and the system is not in lockout, it disables everything to allow normally riding the bike.
- **Alarm Thresholds:** Our system evaluates accelerometer data. If light motion is detected >250mg, it triggers stage 1 warning, while heavy motion >450mg triggers stages 2 warning.

- **Trigger Counter:** If 3 alarms are triggered within a 5 minute window, the system automatically goes to lockout mode.
- **Geofence Check:** When the bike is parked, if the bike is moved out of 10 metres radius from its initial position, it triggers an immediate alert.

Action Layer

- **Log Events:** All temper events are recorded in the EEPROM. It saves timestamp, duration, GPS location etc.
- **Stage 1 and Stage 2 Alarms:** Based on the motion stage arduino activates the buzzer and vibration motor with different patterns.
- **LED Outputs:** Different LEDs provide visual feedback for system status.
- **Kill Switch and SMS:** When critical breaches occur the system engages the ignition relay and sends SMS with location to the bike owner.

Challenges

We faced some challenges like the GSM module won't be able to send sms. Later we understand that GSM and GPS modules both use the same communication protocol so arduino can't communicate with both of them at a time. Then we added some delay in between them to fix the issue. Also, the GSM module's default baud rate was different, we set it to 9600bps. Then, it worked perfectly. Another one is that the GPS module wasn't able to detect location, later we went under the open sky and it successfully detected location.

4. Sustainability & Impact

Sustainability

- **Energy Efficiency:** Power efficient task scheduling and software is used. It economises power by polling for SMS every 15 seconds, and determining GPS geofence coordinates every 30 seconds (only when disarmed and the motorcycle is parked). To further reduce the system's power consumption, it does not poll the accelerometer during "Ride Mode", keeps components such as the buzzer and vibration motor completely off while disarmed, and only writes to the EEPROM memory at the end of a complete tamper event.

- **Material Use & Reusability:** The hardware is based on re-usable and modular components (Arduino Uno, GSM/GPS, relays, sensors). They can be quickly removed, modified and re-used in other IoT projects and thus do not create electronic waste.
- **Environmental Effects:** Our work has an indirect positive environmental impact. By preventing theft and recovering stolen motorcycles (GPS and SMS notifications) the system prevents motorcycles from being stripped for parts. This helps to extend the life of the product, reduce the need for manufacturing new parts and lower waste. Thereby indirectly it reduces the carbon emissions related with production and disposal of the vehicle.

Impact

- **Community & Societal Benefits:** Motorcycle theft is an issue. Our system is an affordable, smart security. With the inclusion of a real-time text message for geofence breach or physical attack, it adds immediate peace of mind for the owner and greater likelihood of recovery, which ultimately leads to increased security.
- **Industrial Benefits:** The project is a very powerful prototype for aftermarket vehicle security. The multi-step alarms, remote telematics (GSM/GPS) and kill-switch functions demonstrate how cheap microcontrollers are able to compete against expensive systems installed by manufacturers.

Future Work

- **Hardware Upgrades:** At a start, a Real-Time Clock (RTC). Currently the EEPROM stores a log of security events which rely on "seconds since boot" to record time (due to the lack of an RTC). This will be used to record the date and time for security events.
- **Scalability & Communication:** A mobile app can be created and moved from SMS to mobile data (GPRS/4G) communication. This will allow real-time tracking of the location on a map user interface rather than Google Maps links via sms.

Limitations

- **Hardware Limitations:** The downfall of the system is the Arduino Uno doesn't have more than one hardware serial port. The GPS and GSM are connected to the Arduino via the SoftwareSerial library. SoftwareSerial can only listen to one device at a time, so the code has to switch between `gpsSerial.listen()` and `gsmSerial.listen()`. In addition, the Arduino has a 1024 byte EEPROM, which can only store a small number of events.
- **Software Limitations:** There are several blocking `delay()` calls, particularly when it comes to sending an SMS, it will wait up to 4 seconds. During these delays the system can't update the state of the system based on the motion data, or update the GPS position. Also, the text-based SMS is susceptible to delays from the GSM provider.

5. Results & Discussion

The ADXL345 accelerator is used in the motion detection system to continuously measure the acceleration variations over all three axes. It measures the overall magnitude of the vectors and compares them to a baseline to identify abnormal movement/vibration. The system assumes suspicious movement when the dynamic acceleration detected is greater than predefined thresholds and responds with a staged alarm. This will enable the system to differentiate between minor disturbance and the possible tampering or theft.

Scenario	Expected Threshold	Actual Threshold	Response Time	Duration	Response
Light Motion(Stage 1)	>250mg	270-400mg	<100ms	3sec	1kHz beep, no vibration
Heavy Motion(Stage 2)	>450mg	530-900mg	<100ms	5sec	2kHz/1kHz beep, vibration

The alarm verification process makes sure that valid motion is detected before it will turn to a full alert. Upon detecting motion, the system validates the intensity with the threshold checks and identifies whether it is a temporary disturbance or a constant movement. The staged alarm is only activated when events have been verified, which minimises false alarms due to small vibrations or environmental disturbances. The system in Stage 1 triggers a continuous buzzer sound at 1kHz and the vibration motor remains off as a gentle warning. Stage 2 is an increase in response to 2kHz + 1kHz alternating buzzer pattern with the vibration motor ON, a high alert to serious tampering or theft attempts.

Output Components	Stage 1	Stage 2
Buzzer	1kHz continuous tone	2kHz+1kHz alternating
Vibration Motor	OFF	ON

The auto-arm and lockout system improves security by automatically checking repeated alarm-triggers and preventing access to the system whenever suspicious activity is observed. In the event that the alarm is activated more than once within a specified time frame, the system puts itself in lockout and prevents the normal operation and turns on the ignition kill switch. All sensor input is disregarded during lockout, and the system can only be reset by a pre-determined series of secret buttons. This not only prevents illegal access but also prevents multiple attempts of tampering with the bike.

EEPROM tamper log is a storage place where records of security events can be stored whenever a motion alarm is activated. Every event logs the time, time duration and the gps position and the system is able to maintain a record of potential mischief.

To determine physical movement of the motorcycle, a magnet on the motorcycle stand is installed and monitored by the reed switch. When the bike is parked and the stand is down, the switch stays open (HIGH). When someone starts riding the motorcycle obviously s/he will lift the stand, the magnet closes the switch (LOW). When this closing is sensed while the system is armed, it is considered as movement and an alarm is released. If the reed switch remains closed it sends an SMS alert with the GPS coordinates and if the switch state is returned to open state it

removes the movement flag. Repeated open/close transitions are properly identified, and the tracking of motion events is guaranteed.

The ride mode feature enables the system to turn off motion-based alarms when the motorcycle is being used legitimately. Ride mode is activated when the key input (D3) is inserted (LOW) and motion detection is disabled to avoid false alarms when the bikes are being ridden normally. When in ride mode, no movement will trigger an alarm, although lockout protection is on, to guard against security. To turn off the ride mode (HIGH) and re-enable full monitoring including motion detection is automatically re-enabled when the key is removed.

The system combines GPS and GSM systems to give a real time tracking and remote communication to ensure improved security. The GPS module constantly receives information on location (latitude and longitude), and this can be used in geofencing and provided as part of the alert in cases of suspicious activity. The GSM module allows the system to send SMS alerts to the authorized user, such as movement, lockout, and regular geofence updates alerts. It also enables receiving commands (e.g., status request), so that the user can remotely track the condition and the position of the motorcycle.

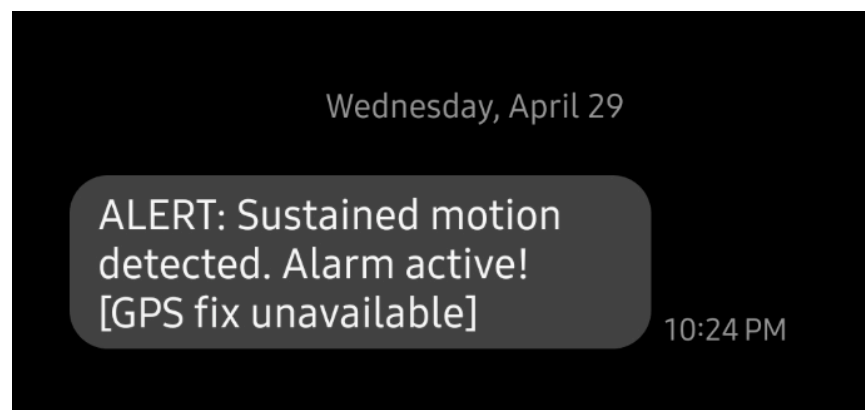


Fig: SMS System Display on Mobile Phones of Vehicle users

Table: Test results of GPS detection subsystem

Event	Time	Duration	Latitude	Longitude
1	5	8s	0.00000	0.00000

2	18	4s	0.00000	0.00000
3	432	11s	23.77272	90.42482
4	792	9s	23.77272	90.42482

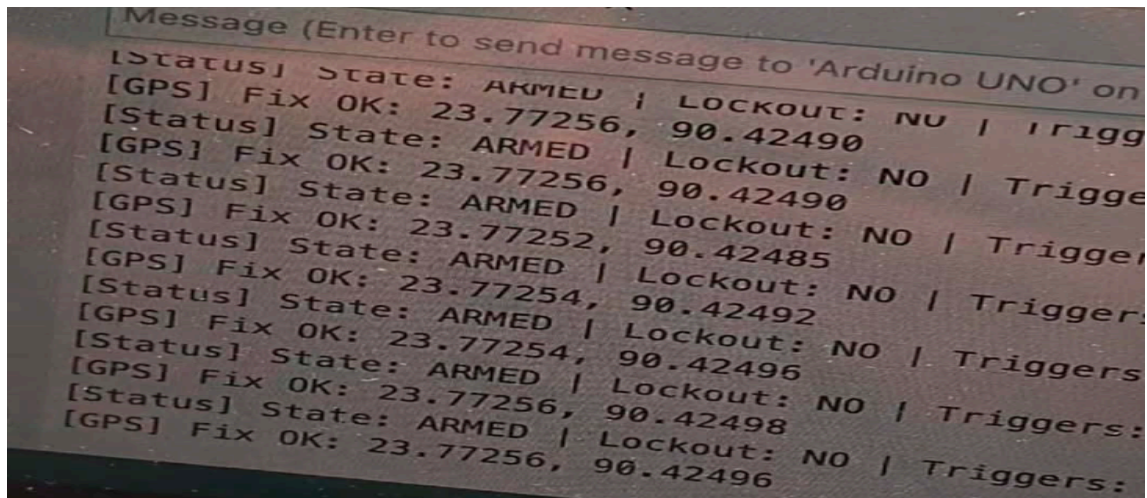


Fig: Test results of GPS detection subsystem

The button control system enables the user to handle and control the security system with a predefined input sequence. Actions like arming/disarming the system, test alarms, and accessing/clearing tamper logs are executed by three buttons. Also, a special set of button presses should be entered during lockout mode to unlock the system, such that only authorized users can retake control.

The system is able to exhibit multi-layer vehicle security with state transitions as events. All the motions, reed detectors, and button-based authentication systems work together to activate the alarms, lockout, SMS, and EEPROM. Repeated motion above threshold during early operation causes the increment of the trigger count and the initiation of staged alarms (beep → loud alarm), then activating the lockout mode after 3 triggers. EEPROM reliably logs every tamper event with a timestamp, duration and GPS coordinates where possible.

Reed switch integration works well to identify bike movement/parking conditions. Once the reed is already closed (movement is detected when armed), GSM SMS alerts are immediately sent out by the system. Upon being opened, it signifies a parked state and forestalls incorrect interpretation of movements. SMS contains GPS data where there is a fix; otherwise it contains fallback no fix.

Properly enabled Ride Mode does not allow motion-based alarms to be used but lockout protection is enabled, allowing safe riding without false detection. The EEPROM logs of tampering are also forensically traceable as they do not change when in ride mode. GSM/GPS/GPRS modules work as anticipated, with GSM offering alert delivery, and GPS offering location tagging when a satellite lock is attained.

6. Conclusion

This project came up with a Smart Bike Security and Tamper Detection System based on the Arduino Uno with an ADXL345 accelerator, reed switch, GPS, and a GSM to overcome the shortcomings of the traditional motorcycle security systems. It achieved a two-stage dual sensor tamper detection with sub-100ms response time, a two-stage alarm system, anti-brute-force lockout, realtime SMS notifications, and tamper logging (EEPROM-based). The system is also operating on an isolated battery and as such, it is not affected by the fact that the main power of the motorcycle is off. Sustainability wise the low power design and reusability of modular components reduce the electronic waste, the theft deterrent and GPS recovery capability of the system indirectly reduces the environmental cost of producing replacement parts. Overall it is a cheap, smart, and energy-saving alternative to the current security systems, and has a great prospect of practical implementation and further development in the future.

7. Contribution

Name	ID	Role(s) Responsibilities	Specific Contributions
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Kaoushik Debnath Joy	22299439	Hardware design and Circuit building	● Selected the required hardware for the project ● Designed the circuit & helped with the software design
Umma Marium Alam	22201096	Hardware design and Circuit building	● Designed the circuit & helped with the software design ● Built the circuit with proper wiring and components.
B M Rauf	22201782	Software design	● Built the core code base on which the whole system runs on ● Implemented the Sms Gsm feature and others.
Azmari Sultana	22201949	Software design & Circuit Building	● Helped build the whole circuit ● Implemented motion sensing, alarm system and logging.

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